INSTRUCTIONAL DESIGN PORTFOLIO

Leyli Nouraei Yeganeh



Instructional designer and researcher

Binghamton, New York

I design learning experiences that make complex ideas easier to understand. My work spans classrooms, teacher development, and research on how people learn with new technologies.

My background in teaching, applied linguistics, and translation shapes how I design instruction. I start by listening to learners, then use their work and feedback to refine the materials.

IN THIS PORTFOLIO

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CONTACT & PROFILES

lnourae1@binghamton.edu Portfolio LinkedIn Google Scholar

DESIGN IN PRACTICE

Areas of expertise

Select an area to see the project, teaching experience, or design guide behind it.



TECHNOLOGY

Generative AI

Turning recorded STEM lessons into visual guidance with STREAM.



TECHNOLOGY

Immersive technologies (VR & AR)

Designing for inclusion in virtual classrooms with Meta-MILE.



DESIGN

UX/UI design

Studying how learners navigate a virtual design studio.



DESIGN

Instructional design

Working from learning objectives to practice and assessment.



EDUCATION

STEM education

Exploring children's emotions during hands-on engineering tasks.



EDUCATION

Teaching

Teaching English, tutoring mathematics and reading, and supporting educators.



EDUCATION

Learning

Choosing how learners practice, receive feedback, and revisit a skill.



EDUCATION

Educational leadership

Studying teacher development in a microelectronics research program.



RESEARCH

Mixed-methods analysis

Reading preschool sequencing scores alongside observations of prompting.

DESIGN IN PRACTICE

Instructional methods

Six frameworks for planning instruction. Each title links to the full method and further reading.



ADDIE Model

Analyze, design, develop, implement, and evaluate.

Pilot an exercise on phone-plan fees, then check whether learners include the fees when comparing two new plans.



Backward Design

Choose the outcome and evidence before planning instruction.

Set the criteria for a scale floor plan, then plan the measuring and drawing practice learners need.



Gagné’s Nine Events of Instruction

Sequence attention, demonstration, guided practice, feedback, and transfer.

Model how axes affect a bar chart. Have learners interpret a fresh chart independently, then revisit the skill with a news graphic.



Merrill’s First Principles of Instruction

Build instruction around a meaningful whole task.

Teachers recall a fraction misconception, watch a modeled response, rehearse with a partner, and adapt it for class.



Universal Design for Learning (UDL)

Offer multiple ways to engage, access information, and show understanding.

Learners justify an estimate in speech, writing, or a labeled sketch. Apply the same reasoning criteria to each response.



Bloom’s Revised Taxonomy

Match the thinking demanded by an assessment to the learning objective.

To assess judgment, ask which of two charts supports an attendance claim and have learners explain their choice.

DESIGN IN PRACTICE / 1 OF 2

Instructional strategies

Each strategy links to a guide. The examples show how a designer might use it in a lesson.



Microlearning

Compare the price per can in two multipacks after a short worked example.



Guest experts and resource speakers

A surveyor explains why a measurement needed checking. Learners decide what to verify on a different plan.



Blended learning

Annotate a misleading graph online, then work in pairs to redraw it in class.



Instructor-led training (ILT)

Model a bus timetable and ask learners to plan an arrival by 8:30. Revisit any column they misread.



Gamification

Unlock a new clue by finding a misleading axis and explaining how it changes a graph's message.



Social and peer learning

Compare 20% off $50 with a $12 coupon. Each partner explains the other's reasoning.



Interactive webinars

Poll teachers on two exit tickets, show sample responses, then invite a revision in the chat.



Hands-on training

Align paper number lines marked in quarters and eighths to explain why 3/4 equals 6/8.



Spaced practice and repetition

Find the median of a fresh data set a day later, a week later, and again later in the term.



Audio learning

Pause a narrated example, find an ambiguous "they" in the transcript, and rewrite the sentence.

DESIGN IN PRACTICE / 2 OF 2

Instructional strategies

Each strategy links to a guide. The examples show how a designer might use it in a lesson.



Storytelling

A café cuts a $20 price by 20%, then raises it by 20%. Predict the final price and explain why it is $19.20.



Online simulations

Change the bounce probability on a virtual pegboard and compare the distributions across runs.



Just-in-time training (JITT)

Place a worked example of percentage change beside the spreadsheet field where learners need it.



Problem-solving

Choose a combination of six-roll and ten-roll packs for 28 guests, weighing cost against leftovers.



Practice drills

After modeling “make ten,” practice 8 + 7 and 9 + 6. Use counters to revisit a missed item.



Scenario-based learning

A child thinks 1/8 exceeds 1/6. Teachers choose a follow-up question and revise it after the child replies.



Personalized and self-guided learning

Skip an already-mastered graph-scale review and use a rubric to check an explanation of a trend.



Experiential learning and evaluation

Give directions for drawing a polygon. Compare the result with the original and revise an ambiguous phrase.



Multimedia learning

Pair a narrated tile animation with a still diagram to connect four rows of six to 4 × 6.

01 / 2025 | FRAMEWORK DESIGN

Meta-MILE

The Future of Education: A Multi-Layered Metaverse Classroom Model for Immersive and Inclusive Learning



Concept illustration of the learning experience.

MY CONTRIBUTION

Conceptual design, writing, editing, visualization, and cross-disciplinary collaboration

CONTEXT

Inclusive virtual learning; conceptual framework development

Inclusive by design

A multi-layered model for immersive learning.

The learning challenge

How can emerging virtual environments become spaces for inclusive learning? Meta-MILE connects immersive technology with instructional design and learner needs.

My contribution

I helped shape the learning framework, wrote and edited research content, and used visualization to communicate the model across disciplines.

What the work demonstrates

Conceptual thinking, clear communication, and instructional design come together to make emerging technology understandable and relevant to a learning experience.

This is a conceptual framework. The portfolio does not report a learner-outcome evaluation for Meta-MILE.

Read the full case study online

02 / 2025 | AI-SUPPORTED STEM CONTENT

STREAM

A Semantic Transformation and Real-Time Educational Adaptation Multimodal Framework in Personalized Virtual Classrooms



Concept illustration of the learning experience.

MY CONTRIBUTION

Design, writing, editing, accessibility thinking, and quality review

CONTEXT

An AI-supported content workflow; offline feasibility pilot

9.2 / 10

Average content-quality rubric score in the pilot.

The learning challenge

Recorded STEM lessons contain valuable information, but learners also need a clear path through it. STREAM transforms recordings into concise, sequenced, and traceable visual guidance.

My contribution

I contributed to workflow design, wrote and edited instructional content, and brought accessibility thinking and quality review to the materials.

What the work demonstrates

The project connects instructional judgment with AI-supported content transformation. Organization, wording, sequencing, and review help the result serve a learner's needs.

The 9.2/10 score describes pilot content quality. It is not a measure of student learning gains or effectiveness at scale.

Read the full case study online

03 / 2025 | USER RESEARCH AND IMMERSIVE LEARNING

Immersive Future

A Case Study of Metaverse in Preparing Students for Career Readiness



Concept illustration of the learning experience.

MY CONTRIBUTION

Study design, analysis, and recommendations for training and support

CONTEXT

Computer-design students; a 20-minute virtual design-studio experience

16 of 28

Students reported increased comfort with the technology.

The learning challenge

Trying unfamiliar technology involves comfort and usability as well as features. This study explored usability, comfort, and perceived career relevance in a brief VR design-studio experience.

My contribution

I contributed to study design, analyzed the student experience, and developed recommendations for preparation, training, and support.

What the work demonstrates

Listening to learners can reveal the guidance and support that need to surround a new technology. Their experiences informed recommendations for a more approachable learning environment.

This is self-reported comfort after a brief experience. It does not establish long-term learning or career-readiness outcomes.

Read the full case study online

04 / 2025 | PRESCHOOL LITERACY & CODING

Stories into sequences

Sequencing Nursery Rhymes Through Early Coding in Preschool Settings



Concept illustration of the screen-free learning experience.

MY CONTRIBUTION

Coauthor; documenting and communicating the instructional approach

CONTEXT

4 children, ages 4 and 5; three lessons in a preschool setting

4 of 4 children

Correctly sequenced all four story images after three lessons.

The learning challenge

We explored how hands-on, screen-free coding could support four preschool children who struggled to sequence familiar stories while introducing basic coding vocabulary.

How the experience worked

The teacher connected Jack and Jill with grids, directional arrows, and story-picture cards. Modeling, guided and paired practice, movement, repeated vocabulary, prompting, and feedback supported the children.

As a coauthor, I helped document and communicate the approach. We examined assessment scores alongside descriptive notes.

What I learned: sequencing skills and coding vocabulary need separate assessment. Future work should examine retention, transfer to unfamiliar stories, and prompting needs.

Read the full case study online

Reported sequencing scores

Learner	Before	After
Child 1	0 / 4	4 / 4
Child 2	1 / 4	4 / 4
Child 3	1 / 4	4 / 4
Child 4	2 / 4	4 / 4

Calculated mean accuracy: **25% to 100%**, a **75-percentage-point** gain.

Two children used materials without prompts; two needed help. One recalled “arrow,” one “forward,” and two recalled no coding terms. Children often called the forward arrow “up.”

Coauthored in The Reading Teacher. Four children, no comparison group, and no longer-term assessment limit conclusions about effectiveness and retention.

05 / SPRING 2024 PROGRAM | 2025 ASEE PAPER

Making room for emotion

Exploring Elementary Students' Emotional States within Engineering Design Tasks in an Afterschool Program (Fundamental)



Concept illustration of the learning experience.

MY CONTRIBUTION

STEM instructor, technology tutor, researcher, and first author

CONTEXT

9 learners in grades 3-5; NSF-supported public-library program in New York

88%

Of reported emotional responses were positive; 12% were negative.

The learning challenge

I designed and taught hands-on engineering with a Binghamton University team while investigating how children experienced success, difficulty, and frustration.

How I supported the experience

Activities included Makey-Makey pianos, robots, drones, light-up cards, and toy modification. I supported troubleshooting, smaller steps, and peer collaboration.

Our team collected visual emotion-tracking sheets and structured interviews. I helped categorize responses and examine patterns across activities.

What we found

Working pianos and illuminated cards often prompted excitement and pride. Drone assembly and toy modification could prompt confusion or frustration when parts were unfamiliar or designs failed.

Children described seeking peer help, trying alternatives, and feeling proud when a project worked.

What I learned

Treat emotions as instructional feedback, balancing visible progress with meaningful challenge and timely support.

The 88% figure describes responses, not students who improved. The small, self-selected sample was exploratory; achievement, engagement, and resilience gains were not directly measured.

Read the full case study online

06 / SUMMER 2026 | SIX-WEEK RET PROGRAM

Research that reshapes teaching

Examining Shifts in Secondary STEM Teachers' Understanding of Engineering Through a Microelectronics Research Experience for Teachers (RET) Program



Concept illustration of the learning experience.

MY CONTRIBUTION

Program administrator, researcher, and data analyst

CONTEXT

8 secondary STEM teachers; NSF-funded program at Binghamton's Watson College

10 of 11

Constructs had higher average coding scores after the program.

The research question

How do authentic microelectronics and semiconductor research experiences relate to teachers' understanding of engineering?

My contribution

I supported interdisciplinary coordination and analyzed pre/post questionnaires covering 11 constructs. Two researchers coded independently on a 0-4 scale and resolved differences through discussion.

I examined descriptive statistics alongside changes in teachers' language, specificity, and laboratory or industry examples.

What I learned

Reading explanations alongside scores helped interpret uneven results. Future evaluation should follow up on retention and classroom practice.

Read the full case study online

Selected construct scores

Pre / post means on the 0-4 coding scale.

Construct	Pre	Post
Social / collaborative	1.75	2.62
Creativity / imagination	1.86	2.29
Criteria / constraints	1.88	2.38
Role of failure	2.12	2.62
Empirical testing	2.29	2.00

Written responses became more specific and grounded in experience. These descriptive findings from eight teachers do not establish causality or later classroom impact.

07 / SPRING AND FALL 2025 | TUTORING

Reading through what matters

Using Generative AI to Support Reading Engagement in an Adolescent with Reading Disabilities



Concept illustration of the learning experience.

MY CONTRIBUTION

Mathematics tutor and instructional-material designer

CONTEXT

An adolescent learner; Windsor High School, under Binghamton University supervision

Interest-led learning

Observed progress in reading and willingness to attempt tasks.

The learning challenge

Written mathematics problems can create reading barriers. Within an NSF-supported team project, I designed personalized activities connecting students' interests with the reading demands of problem-solving.

How I designed the materials

I used generative AI to help draft interest-based materials, then adapted them to the learner. Stories, games, drawing, and puzzles offered varied ways to participate.

I observed reading accuracy, fluency, participation, and willingness to try. Biweekly meetings with teammates and our supervisor helped refine instruction.

What I observed

Tutoring observations indicated improvements in reading accuracy and fluency, stronger engagement with personalized materials, and greater motivation to attempt mathematical problems.

What I learned

Connecting literacy and mathematics through interests can support participation. Future evaluation should document separate baseline and final measures for accuracy, fluency, engagement, and motivation.

No numerical reading improvement is claimed without verified assessment details. The evidence does not isolate AI's contribution from personalization, tutoring, or other strategies.

Read the full case study online

08 / FALL 2023 STUDY | 2026 PUBLICATION

Writing with intention

Integrating Self-Regulated Learning in Flipped Learning to Enhance Argumentative Writing with Digital Tools



Concept illustration of the learning experience.

MY CONTRIBUTION

First and corresponding author; research and instructional-design collaborator

CONTEXT

240 medical students with intermediate English; University of Kashan, Iran; 16 weeks

+1.92 points

Adjusted writing-score difference; p < .001; eta-squared = .14.

Connect tools to learning purposes

We compared flipped instruction with and without explicit self-regulation support. Google Docs supported goals and collaborative drafting; EdPuzzle, pre-class checks; Padlet, reflection; and Quizizz, practice.

Evaluate the writing process

Two raters, blind to group, assessed essays for claims, qualifiers, evidence, and coherence. We also used a 41-item SRL questionnaire and baseline-adjusted analysis.

What I learned

Digital tools need a clear instructional purpose. In the supported group, claim scores rose from 1.52 to 2.75 and qualifiers from 1.46 to 3.19; about 15% of essays still lacked appropriate qualifiers.

Reported group mean scores

Measure / group	Pre	Post
Writing / SRL support	8.73	16.80
Writing / comparison	8.50	14.78
SRL / supported	104.5	166.7
SRL / comparison	103.2	116.1

Comparison: flipped learning without explicit SRL support.

The findings interpret assessed writing outcomes, not a separate interview or thematic analysis. The study was published in the Journal of Modern Research in English Language Studies in 2026.

Read the full case study online

HOW I WORK

Curious first. Clear always.

A thoughtful practice, with room to listen, test, and change direction as we learn.



Content with intention

Sequenced learning, curriculum design, and writing with clarity and consistency.



Research with purpose

Audience analysis, mixed methods, and data-informed content decisions.



Technology with care

AI-supported workflows, prototyping, and immersive learning experiences.



Better work, together

Cross-disciplinary collaboration, stakeholder communication, and shared delivery.

A PROCESS THAT LEARNS

01

Understand people

Listen to goals, needs, context, and difficult moments.

02

Make it clear

Shape complex ideas into thoughtful sequences and precise language.

03

Make it tangible

Build guides, interactive lessons, and prototypes to explore.

04

Keep learning

Use feedback, engagement data, and research to refine the experience.

TOOLS & METHODS


n8n


Lovable


Figma


Docker


ChatGPT


Gemini


EdPuzzle


Moodle


Canvas


Google Workspace


Microsoft Office


Padlet


SPSS


Python


Excel


VR / AR


Mixed-methods research


Accessibility


Peer review

HOW I WORK / 1 OF 2

Tools & methods

Each tool or method links to a guide. The examples show a specific use in learning design or research.



n8n

Route a draft lesson to a review queue and notify the editor, who decides when it is ready to publish.



Lovable

Prototype a learner's path from a scenario to a decision and feedback. Ask a colleague to try it.



Figma

Test whether a learner can find feedback and revise an answer in a clickable "try again" prototype.



Docker

Package a coding exercise with its libraries so learners can start the workshop with the same setup.



ChatGPT

Draft table-based and graph-based explanations of rate of change. Review the mathematics and reading level.



Gemini

Draft questions about a graph, then verify its axes, units, and answer choices before using them.



EdPuzzle

Pause a video before the result and ask for a prediction. Use the responses to decide which step to revisit.



Moodle

Link feedback on a missed quiz question to the worked example that explains the step.



Canvas

Put a design brief, sample submission, and rubric in one module so learners can check their drafts.



Google Workspace

Review a storyboard in Docs and track open revision decisions in a shared Sheet.

HOW I WORK / 2 OF 2

Tools & methods

Each tool or method links to a guide. The examples show a specific use in learning design or research.



Microsoft Office

Check a facilitator guide in Word against its PowerPoint slides so both give the same activity instructions.



Padlet

Have learners post two different solution sketches and compare the reasoning behind them.



SPSS

Match pre- and post-assessment scores, then summarize the change for the learners with paired results.



Python

Plot distance against time. Let learners change a value and predict how the slope will change.



Excel

Count missed responses by question in a pivot table, then review the most missed item against its objective.



VR / AR

Rehearse a procedure in a virtual lab or use AR labels to identify the same equipment on a physical bench.



Mixed-methods research

Compare scores with field notes. All four preschoolers sequenced the story, while two still needed prompts to use the materials.



Accessibility

Complete a lesson using only a keyboard. Check focus visibility and whether graph descriptions convey the relationship being taught.



Peer review

Have a reviewer complete a task using only the learner instructions. Revise steps they interpret differently.

EXPERIENCE

Different settings.
The same curiosity.

From classrooms to research teams, I help people navigate complex ideas, build skills, and find their next step.



2023-PRESENT

AI and Immersive Learning Programs

Instructional Design Researcher

Contribute to conceptual design, writing, visualization, and evaluation across AI-supported content, immersive learning, and instructional frameworks.



AUG 2025-MAY 2026

Windsor Middle School

Mathematics & Reading Tutor

Delivered targeted small-group instruction, adapting materials, explanations, and pacing in response to demonstrated understanding.



MAY-AUG 2025

Binghamton University

Instructional Technology Assistant

Developed train-the-trainer materials and clear demonstrations of assistive and special education technologies, with attention to accessibility and digital equity.



AUG-DEC 2024

Binghamton University

STEM Program Instructor

Designed hands-on and gamified K-12 STEM content; adapted pacing and sequencing to learner needs and used engagement data to revise instruction.



SEP 2017-JUN 2022

University of Qom

Lecturer

Developed curriculum materials, assessments, and testing tools. Sequenced complex concepts and reviewed writing for clarity, style, accuracy, and consistency.



JUN 2020-JUN 2022

Arad Branding

Recruiting Specialist

Managed concurrent full-cycle hiring pipelines, coordinated timelines and requirements, and communicated across departments and stakeholders.



APR 2012-MAR 2016

Motarjem Institute

English Language Instructor, K-12

Created audience-appropriate instructional content, flipped-learning video lessons, and self-regulated learning materials.

RESEARCH & WRITING

Thinking beyond the project.

Selected publications on learning, language, and the possibilities of emerging technology.



2026

Integrating Self-Regulated Learning in Flipped Learning to Enhance Argumentative Writing with Digital Tools

Journal of Modern Research in English Language Studies



2025

The Future of Education: A Multi-Layered Metaverse Classroom Model for Immersive and Inclusive Learning

Future Internet | MDPI



2025

STREAM: A Semantic Transformation and Real-Time Educational Adaptation Multimodal Framework in Personalized Virtual Classrooms

Future Internet | MDPI



2025

Exploring Elementary Students' Emotional States within Engineering Design Tasks in an Afterschool Program (Fundamental)

ASEE paper



2025

Sequencing Nursery Rhymes Through Early Coding in Preschool Settings

The Reading Teacher | Wiley / ILA



2025

Immersive Future: A Case Study of Metaverse in Preparing Students for Career Readiness

IEEE MetaCom 2025 | Conference proceedings

EARLIER WORK IN APPLIED LINGUISTICS

2016 | A Comparison of the Effects of Dictogloss and Oral Dialogue Journal Techniques on Iranian EFL Learners' Acquisition of Request Speech Act

2016 | An Analysis of Pragmatic Competence in 2013 Presidential Election Candidates of Iran

Let's make learning mean more.

lnourae1@binghamton.edu

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